

ARRAYS

Q7) Sum Of Matrix Submatrix Elements

DS	Session	Arrays	Question Information	Level 1	Challenge 27
Problem Description: Nirobi have given a matrix C of size N x M to Rio. Also Rio are given position of submatrix as X1, Y1 and X2, Y2 inside the matrix. Now Rio needs to find the sum of all elements inside that submatrix. Can you help Rio in completing the task assigned by Nirobi. Constraints: $1 \leq T \leq 15$ $1 \leq N, M \leq 103$ $1 \leq C[N][M] \leq 106$ $1 \leq X1, Y1, X2, Y2 \leq M$ Input Format: The first line of input contains an integer T denoting the number of test cases. The first line of each test case is n and m,n is the number of rows and m is the number of columns. The second line of each test case contains C[N][M]. The third line contains four value of X1, Y1, X2, Y2. X1, Y1 is the top left cell and X2, Y2 is the bottom right cell.					

CODE:

```
C main.c
1  #include <stdio.h>
2
3  void solve() {
4      int n, m;
5      if (scanf("%d %d", &n, &m) != 2) return;
6
7      long long mat[n + 1][m + 1];
8      for (int i = 1; i <= n; i++) {
9          for (int j = 1; j <= m; j++) {
10             scanf("%lld", &mat[i][j]);
11         }
12     }
13
14     int x1, y1, x2, y2;
15     scanf("%d %d %d %d", &x1, &y1, &x2, &y2);
16
17     long long sum = 0;
18     for (int i = x1; i <= x2; i++) {
19         for (int j = y1; j <= y2; j++) {
20             sum += mat[i][j];
21         }
22     }
23
24     printf("%lld\n", sum);
25 }
26
27 int main(void) {
28     int t;
29     if (scanf("%d", &t) == 1) {
30         while (t--) {
31             solve();
32         }
33     }
34     return 0;
35 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
1
3 3
1 2 3
4 5 6
7 8 9
1 1 2 2
12
```